DSCI9001 Foundations of Data Practice

Assignment 2: Reproducible Analysis (FICTIONAL EXAMPLE)

Overview

This assignment is worth **30%** of your final grade and is marked out of **100**. It is due **Friday 13 November 2026, 11:59 pm AEDT** via the Moodle assignment box (Turnitin). This is an individual assignment.

The assignment has two questions. Question 1 is a written discussion; Question 2 is a small programming task with a short write-up.

Question 1: Reproducibility practices (50 marks, approximately 300 words)

Critically discuss **two** practices that improve the reproducibility of data analyses. For each practice, explain what problem it addresses and evaluate at least one limitation. You must draw on the set reading and the Week 3 lecture.

Question 2: Smoothing a sensor series (50 marks, approximately 200 words plus code)

Using the starter code, implement a centred moving-average smoother and apply it to the supplied sensor series (`sensor.csv`). Report the root-mean-square error (RMSE) between the smoothed series and the reference signal for window sizes 3 and 7, and interpret the result. You should include at least one figure comparing the raw, smoothed and reference series. Your code must pass the supplied tests with `python3 -m pytest`.

Requirements

- You must use Harvard referencing.
- Word limits are approximate.
- Write in Australian English.
- Submit a single PDF named `DSCI9001_A2_<studentID>.pdf`, plus a zip file of your code.
- Your code must run with Python 3.10 or later. You may use numpy and matplotlib.
- You must include an "AI use statement" section at the end of your submission.

Use of generative AI

You may use generative AI tools for brainstorming, feedback on your drafts and code assistance. You must not submit AI-generated text as your own without acknowledgement. You must acknowledge any use in an "AI use statement" describing the tools used and how you used them.